

D I C E

THE HARDWARE-BACKED VRF ORACLE ON SOLANA

Trust the device, not the daemon.

DICE is the only randomness oracle on Solana whose entropy is rooted in physical, attested hardware. Five ESP32-S3 devices commit-reveal across mTLS, finalised on-chain by an Anchor program. Every random bit traces back to a piece of silicon you can hold in your hand.

• devnet live

5 ESP32-S3 nodes

commit-reveal v2

0.002 SOL / request

Every randomness oracle on Solana asks you to trust a daemon.

- **Switchboard, ORAO, Kamui** — software-only DON. The "decentralisation" is N anonymous Linux processes you can't audit.
- **MagicBlock Ephemeral VRF** — clever, fast, but still a software guarantee on top of a software rollup.
- **Chainlink VRF** — not Solana-native; same software-DON pattern.
- **The oracle problem is 2,000 years old.** The Delphi priestesses had the same job. They were also the same single point of failure.

WHAT EVERY DAPP ACTUALLY WANTS

A **physical chain of custody** for entropy: from a hardware RNG, through a tamper-evident path, to an on-chain commitment — auditable end to end.

"The oracle problem refers to the inability of an agent to know if the information coming from an oracle is authentic and unbiased."

— *Two thousand years of the oracle problem · Insights from Ancient Delphi*

A randomness oracle rooted in silicon.

01 · PHYSICAL ENTROPY

Five ESP32-S3 devices, each with a hardware RNG and a secp256k1 device key burned into NVS. Entropy is generated **on the chip**, not in a process.

02 · COMMIT-REVEAL ACROSS MTLS

Each device sends a SHA-256 commit, then reveals the entropy + ECDSA signature only after all commits are bound. Front-running is impossible by construction.

03 · ON-CHAIN FINALITY

An Anchor program verifies each signature, hashes the entropies, and writes the result + per-node payouts atomically. Devnet program ID

78Qv6cyKkRZN2YngiLSSBCe2iyRc6jgtCs3incCaMRcv.

THE DICE CHAIN OF CUSTODY

LAYER 1

ESP32-S3 ×5



LAYER 2

mTLS coord



LAYER 3

Solana program

TRACTION · THE HEADLINE NUMBER

98.5%

pass rate, **985 of 1,000** sequential v2 rounds

Real ESP32-S3 hardware · Solana devnet · Helius RPC · 2026

7.79s

AVG ROUND

9.64s

P95

5

PHYSICAL NODES

229

UNIT TESTS

• A4 STRESS TEST PASSING

• HARDWARE PAYOUT BINDING LIVE

• STREAMING VRF FEED LIVE

⚙ MAINNET PENDING

A request, end to end, in one slide.

- **1** • dApp calls `request_randomness_auto` on a pre-funded DiceChannel PDA.
- **2** • Coordinator selects 4 of 5 active devices via on-chain `select_nodes` + `SlotHashes` randomness.
- **3** • Each device generates entropy on its RNG, signs `SHA-256(entropy)`, sends the commit over mTLS.
- **4** • After all commits land on-chain, devices reveal the entropy + ECDSA signature.
- **5** • Program verifies each signature, hashes the entropies, writes the result, splits the fee 70/20/10 in one atomic TX.

WHY EACH LAYER MATTERS

- **Hardware:** entropy never leaves the chip in plaintext until the reveal phase.
- **mTLS:** the wire is encrypted; the device cert binds identity to silicon.
- **Commit-reveal:** the coordinator can't see entropy before the commits are sealed on-chain.
- **On-chain finality:** every reveal is verified by the Solana program; the result is just `SHA-256(e1 || e2 || e3 || e4)`.

Bias-free as long as **at least one** device honestly contributes entropy — not "as long as $\frac{2}{3}$ of nodes are honest."

Everyone else is software.

PROVIDER	ROOT OF TRUST	LATENCY	HACKATHON / SHIPPED	COST NARRATIVE
DICE us	• hardware (ESP32-S3 + secp256k1 in NVS)	7.8 s today → ~5 s post v7.5	Devnet live · 5 real devices	Flat 0.002 SOL · 70 % to operators
Switchboard VRF	software DON	~5 s	Mainnet · years of integrations	Variable; pull-based
ORAO VRF	software DON	~3-6 s	Mainnet	Variable
Kamui	software DON	n/a	Breakout 2025-04	"Cheapest VRF on Solana"
MagicBlock Ephemeral VRF	software, ephemeral rollup	sub-second (rollup)	Public beta · 2025	Embedded in their stack
WAIT (VDF)	software, MinRoot+zkSNARK	seconds	Cypherpunk 2025-09	Different primitive

Source: Colosseum builder corpus (search/projects), Colosseum Codex archive entry on MagicBlock Ephemeral VRF, public docs.

The crypto isn't the moat. The fleet is.

OPERATIONAL PKI

Private step-ca, secp256r1 mTLS certs + secp256k1 device identity, split-key NVS provisioning. Years of cert hygiene to copy.

HARDWARE SUPPLY CHAIN

A bonded device manufacturer + provisioning pipeline. New entrants can't ship a 5-node network in a weekend.

VERIFIABLE BOOT

Bounded firmware monitor, signed binaries, factory-reset via PKI revocation. Every device is a known-good silicon block.

AUDIT-GRADE STORY

A regulator can trace any random bit back to a specific board, a specific NVS, a specific cert. No software DON can say that.

CYPHERPUNK CREDIBILITY

Closest analogs in the corpus are **TEENet**, **ASSAP**, **WeTEE**, **SUMOTEX**. Hardware-rooted trust is a recognised movement.

STREAMING PRIMITIVE

RandomnessFeed PDA + crank — passive subscribers (e.g. **pulse**) read randomness with zero round-trip. No competitor has this shape.

Per-request economics that pencil for an operator.

FEE PER ROUND

0.002 SOL

≈ \$0.30 at \$150 SOL · charged per request_randomness_auto

SPLIT (ATOMIC, ON-CHAIN)

70%

NODE OPERATORS

20%

TREASURY

10%

RESERVE

OPERATOR UNIT ECONOMICS (ILLUSTRATIVE)

- Hardware cost: ~\$30 per ESP32-S3 dev kit + cert provisioning
- Per round, 4 of 5 nodes earn **0.0014 SOL** ÷ 4 = **0.00035 SOL** each
- 1,000 rounds / day = **0.35 SOL / device / day** ≈ \$52
- Hardware payback: **under 1 day** at current devnet throughput

WHY THIS WORKS AS A FLYWHEEL

More devices → more parallel rounds → lower latency → more dApp demand → more fees → more operator yield → more devices. Same shape as **DeCharge**, **Kiko**, **BlockMesh** — DePIN winners that already proved the pattern at Colosseum.

First-call design partners, all on Solana, all need a VRF.

IGAMING · ALREADY SHIPPING

- **ThroneX** · Renaissance + Radar · zero-sum on-chain iGaming
- **Gene Lotto** · Cypherpunk 2025 · ephemeral-rollup lottery

LOTTERY / RAFFLE APPS

- **Solot**
- **Solana Mini Lottery**
- **SOL Raffle**
- **soltice** · Jogojogo

IN-HOUSE DAPPS (ALREADY WIRED)

- **coin-toss** · v2 callback E2E live
- **pulse** · streaming VRF subscriber
- **dice-roll** · **lucky-wheel** · **prediction-market**

THE PLAY

Free integration for the first 5 design-partners + their logos on the explorer. We write the migration code from Switchboard ourselves. Each integration becomes a mainnet TX we can tweet.

We do not compete on cost. We compete on truth.

- ~~"Cheapest VRF on Solana"~~ — Kamui already owns this lane.
- ~~"Lowest latency VRF"~~ — MagicBlock owns this lane via ephemeral rollups.
- ~~"Most integrations"~~ — Switchboard has years of head-start.
- **"The only randomness with a hardware root of trust"** — nobody owns this lane.
- **"Audit-grade VRF for regulated gambling"** — the buyer who actually pays a premium.
- **"Cypherpunk randomness"** — physical, anti-trust, verifiable. The movement is named after this.

THE ONE-LINER

DICE is the only Solana VRF where every random bit traces back to a piece of silicon you can hold.

WHY NOW

DePIN is **the** Colosseum narrative — Kiko, DeCharge, BlockMesh, GEN are all winners. Hardware-rooted trust is the next leg (TEENet, ASSAP, SUMOTEX). DICE is the first to apply that pattern to randomness.

From devnet headline to mainnet logo.

WEEK 1 · ENGINEERING

- v7.5 ALT-bundle (commits + reveals in one TX)
- Avg latency **9 s** → **~5 s**
- L3-lite bench published

WEEK 2 · PRODUCTION

- Mainnet program deploy
- Coordinator on Railway w/ TCP proxy for mTLS
- HSM-protect coordinator key

WEEK 3 · NARRATIVE

- "Physical attestation > software DON" thesis post
- Public explorer with live device map
- Operator earnings calculator

WEEK 4 · PARTNERS

- Cold outreach: Solot · SOL Raffle · ThroneX
- First mainnet integration TX (tweetable)
- Cypherpunk cohort application

1

MAINNET PROGRAM

3

DESIGN PARTNERS

5s

AVG ROUND TARGET

1

CYPHERPUNK APP

THE ASK

If randomness should be physical, work with us.

DESIGN PARTNERS

Free integration + explorer logo for the first 5 dApps that ship a mainnet round on DICE.

CAPITAL

Pre-seed for hardware fleet expansion (5 → 50 devices), mainnet ops, second engineer, audit budget.

DISTRIBUTION

Cypherpunk-cohort backing, Solana DevRel intros, regulated-gaming auditors who care about provenance.

D I C E

HariHaran · DICE Labs
github · DICE-vrf · 2026

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